Numerical Modelling and Machine Learning Prediction of Single-Particle Crushing Mechanics and Fractal Gradation of Recycled Aggregates

A Project Report submitted by
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in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B.Tech)



Indian Institute of Technology Jodhpur
Department of Civil and Infrastructure Engineering
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DECLARATION

I hereby declare that the work presented in this Project Report titled **Numerical Modelling and Machine Learning Prediction of Single-Particle Crushing Mechanics and Fractal Gradation of Recycled Aggregates**, submitted to the Indian Institute of Technology Jodhpur in partial fulfilment of the requirements for the award of the degree of B.Tech., is a bonafide record of the research work carried out under the supervision of **Dr. Tekcham Gishan Singh** of the respective Supervisor. The contents of this Project Report in full or in parts, have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

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CERTIFICATE

This is to certify that the Project Report titled **Numerical Modelling and Machine Learning Prediction of Single-Particle Crushing Mechanics and Fractal Gradation of Recycled Aggregates**, submitted by **Ayush Kumar (B23CI0xx)** to the Indian Institute of Technology Jodhpur for the award of the degree of B Tech, is a bonafide record of the research work done by him under my supervision. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

Recycled aggregate (RA) derived from crushed concrete, brick, and demolition mortar is widely recognized as a sustainable alternative to depleting virgin mineral quarry stone in civil infrastructure. Incorporating recycled aggregates into highway subgrades, structural concrete, and railway ballast supports circular construction and minimizes construction and demolition (C&D) waste disposal. However, their structural utilization is severely restricted by irregular 3D morphology, residual micro-cracking, and pronounced vulnerability to single-particle crushing under compaction and vehicular loading. Standard laboratory evaluation using Universal Testing Machines (UTM) and mechanical sieve shakers is destructive, labor-intensive (taking 16–24 hours per sample batch), hazardous due to respirable crystalline silica dust, and incapable of mapping continuous variations in particle geometry.

In this study, the single-particle crushing mechanics and post-breakage particle size distribution of recycled aggregates are investigated using a physics-informed machine learning (PIML) framework. Three demolition materials are evaluated: Recycled Concrete (RC), Recycled Brick (RB), and Recycled Mortar (RM). Empirical Weibull brittle fracture size-scaling parameters (Liang et al., 2025) are coupled with 3D sphericity-dependent strength and non-linear energy power laws (Cai et al., 2024), generating a comprehensive benchmark dataset of N = 3,600 individual particles spanning diameters from 4.75 mm to 37.5 mm. Six regression models (Ridge Polynomial, Random Forest, XGBoost, LightGBM, CatBoost, and Physics-Informed MLP) are trained and benchmarked across crushing strength, peak rupture force, crushing energy, and post-breakage fractal dimension. The numerical results are validated against published experimental datasets. Furthermore, an interactive virtual testing suite is deployed in Streamlit to provide practicing engineers with instantaneous, non-destructive particle characterization.

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1. INTRODUCTION

Aggregates constitute the primary granular constituent in civil engineering construction, representing over 70% of concrete volume and almost 100% of unbound highway subgrades, sub-base layers, and railway ballast. The worldwide consumption of natural mineral aggregates exceeds 50 billion tonnes per year, causing widespread ecological degradation including riverbank erosion, groundwater table lowering, and depletion of natural stone quarries. In parallel, global construction and demolition (C&D) waste generation now exceeds 3.0 billion tonnes annually. Discarding this debris into landfills consumes scarce land and represents an inefficient linear economic model. Repurposing C&D waste into Recycled Aggregates (RA) offers a vital closed-loop circular economy pathway.

Despite their environmental benefits, the broader commercial adoption of recycled aggregates in load-bearing structural concrete and pavement bases remains limited. Unlike homogeneous virgin stone (basalt, granite, limestone), recycled aggregates are highly heterogeneous, comprising three primary fractions: Recycled Concrete (RC), Recycled Brick (RB), and Recycled Mortar (RM). Pre-existing microcracks induced during service life and explosive hydraulic demolition impart elevated porosity (water absorption commonly 5–15%), reduced stiffness, and high susceptibility to particle crushing under roller compaction and vehicular wheel traffic. Particle degradation alters the designed Particle Size Distribution (PSD), leading to loss of bearing capacity and structural rutting.

In current geotechnical practice, single-particle crushing resistance is characterized by destructive diametral compression tests inside a Universal Testing Machine (UTM), followed by mechanical sieve analysis. However, this testing regime is destructive, labor-intensive (requiring 16–24 hours per batch), hazardous due to airborne silica dust inhalation, and unable to explore continuous multi-dimensional geometric spaces. To overcome these challenges, this study establishes a Physics-Informed Machine Learning (PIML) framework. By embedding classical geomechanical formulations—specifically Hiramatsu-Oka tensile rupture, Bagheri gauge diameter, Weibull statistical weakest-link size scaling, and Tyler-Wheatcraft fractal fragmentation—into multi-task machine learning architectures, single-particle mechanical behavior and complete post-crushing sieve curves are predicted instantaneously without material destruction.

2. LITERATURE SURVEY

The mechanical crushing of quasi-brittle mineral particles is governed by Griffith's brittle fracture theory (Griffith, 1921), wherein tensile failure initiates at critical microstructural flaws and pores. In irregular grains subjected to point compressive loading between flat platens, Hiramatsu and Oka (1966) derived the classical relationship converting peak compressive failure force F_f to nominal tensile splitting strength σ_f using effective diameter d_0: $\sigma_f = 0.9F_f/d_0^2$. Bagheri et al. (2015) established that surface roughness asperities reduce effective platen gauge contact width relative to volumetric equivalent spherical diameter d_{eq}, formulating the roughness correction: $d_0 = 0.887d_{eq} = 0.887(L \cdot I \cdot S)^{1/3}$, where L, I, and S are orthogonal principal dimensions. This formulation was confirmed and adopted by Liang et al. (2025) for recycled demolition waste particles.

Particle size exerts a pronounced effect on crushing strength. Weibull (1951) weakest-link theory demonstrates that larger material volumes possess a higher cumulative probability of containing catastrophic Griffith flaws, resulting in reduced nominal strength: $P_s(\sigma) = \exp[-(\sigma/\sigma_{f0})^m]$, where σ_{f0} is characteristic strength and m is Weibull modulus. Liang et al. (2025) experimentally tested single-particle crushing of construction waste across

three sieve brackets (4.75–10 mm, 10–20 mm, and 20–25 mm) for RC, RB, and RM. Their results confirmed that characteristic strength systematically decreases with increasing particle size, with RC exhibiting low Weibull moduli ($m = 1.25$–1.58) due to composite heterogeneity between natural stone cores and adhered mortar.

Beyond size, 3D grain morphology strongly dictates stress concentrations. Cai et al. (2024) used 3D optical scanning and DEM simulations to quantify the effect of true sphericity $\Phi = \sqrt[3]{(I \cdot S)/L^2}$ on recycled aggregate crushing. At a reference size ($d_{ref} = 25extmm$), crushing strength follows a strict power law: $\sigma_f = p \cdot \Phi^q$. Furthermore, crushing energy E_f exhibits two distinct regimes separated at $\Phi = 0.80$: a power-law regime for angular grains ($\Phi < 0.80$) and an exponential saturation regime for compact grains ($\Phi \geq 0.80$). Upon rupture, fragments follow scale-invariant fractal distributions (Tyler & Wheatcraft, 1992): $F(d) = (d/d_0)^{3-D}imes100\%$, where fractal dimension D reflects comminution intensity. Recent machine learning applications in granular geomechanics (Wang et al., 2021; Oskooei et al., 2021; Miao et al., 2024) demonstrate high potential for predicting material strength. However, no prior study has unified multi-scale Weibull size scaling with continuous 3D sphericity to simultaneously predict crushing strength, peak force, energy, and full post-breakage sieve gradation.

3. PROBLEM DEFINITION & OBJECTIVES

3.1 Problem Definition

This study addresses the unpredictability and experimental testing bottlenecks associated with recycled aggregate crushing mechanics. The investigation encompasses three primary demolition waste materials: Recycled Concrete (RC), Recycled Brick (RB), and Recycled Mortar (RM) spanning equivalent diameters from 4.75 mm to 37.5 mm. A total of 3,600 physics-coupled samples are synthesized and six machine learning regression architectures alongside three classifiers are benchmarked. The models predict single-particle crushing strength, failure force, crushing energy, and complete post-breakage fractal gradation. Results are quantitatively validated against published experimental datasets to evaluate model fidelity and establish an automated virtual testing platform.

3.2 Objectives

The primary technical objectives of this B.Tech Capstone Project are:

- To synthesize a balanced, physics-informed benchmark dataset of $N = 3,600$ single particles coupling Weibull size scaling and 3D sphericity power laws.
- To develop, tune, and evaluate six multi-task regression architectures (Ridge Polynomial, Random Forest, XGBoost, LightGBM, CatBoost, and Physics-Informed MLP).
- To predict single-particle crushing strength (σ_f), peak failure load (F_f), crushing energy (E_f), and post-breakage fractal dimension (D).
- To implement multi-class classification for predicting discrete breakage failure modes (Mode I Splitting, Mode II Corner Rupture, Mode III Fragmentation).
- To elucidate governing solid mechanics and verify thermodynamic consistency using game-theoretic SHAP and 2D partial dependence analyses.
- To quantitatively validate model accuracy against published laboratory benchmark data from Liang et al. (2025) and Cai et al. (2024).

- To deploy an interactive Streamlit engineering software suite featuring a Virtual UTM, an Inverse Design Scaling Simulator, and an auxiliary Computer Vision scanner.

4. METHODOLOGY

4.1 Physics-Coupled Dataset Generation

4.1.1 Material Properties and Grain Morphology

Baseline physical and micromechanical properties for the three C&D waste lithologies were mined from published laboratory literature (Liang et al., 2025 Table 1; Cai et al., 2024 Section 3.2), as compiled in Table 1.

Table 1. Material and micromechanical properties of modeled C&D waste lithologies

Material Property	Recycled Concrete (RC)	Recycled Brick (RB)	Recycled Mortar (RM)	Literature Reference
Apparent Density (g/cm³)	2.217	2.140	2.028	Liang et al. (2025)
Bulk Density (g/cm³)	1.296	1.063	1.113	Liang et al. (2025)
Water Absorption (%)	5.32	14.41	10.80	Liang et al. (2025)
Elastic Modulus (GPa)	15.2	8.6	9.1	Cai et al. (2024)
Poisson's Ratio	0.20	0.20	0.20	Cai et al. (2024)

Particles were generated across three standard sieve fractions: 4.75–9.5 mm, 9.5–19.0 mm, and 19.0–37.5 mm. Tri-axial dimensions (L, I, S) were sampled from truncated normal distributions centered on Cai et al. (2024) empirical means $(CV_L = 12\%, CV_I = 10\%, CV_S = 13\%)$, sorted descending to guarantee $L \geq I \geq S > 0$. Shape indices were computed: Elongation Index $EI = I/L$, Flatness Index $FI = S/I$, True Sphericity $\Phi = \sqrt[3]{(I \cdot S)/L^2}$, and Domokos Shape Factor S_f. Zingg classification was mapped using the 0.67 boundary threshold, distributing particles realistically across Disk, Block, Blade, and Bar morphologies (Figure 4.1).

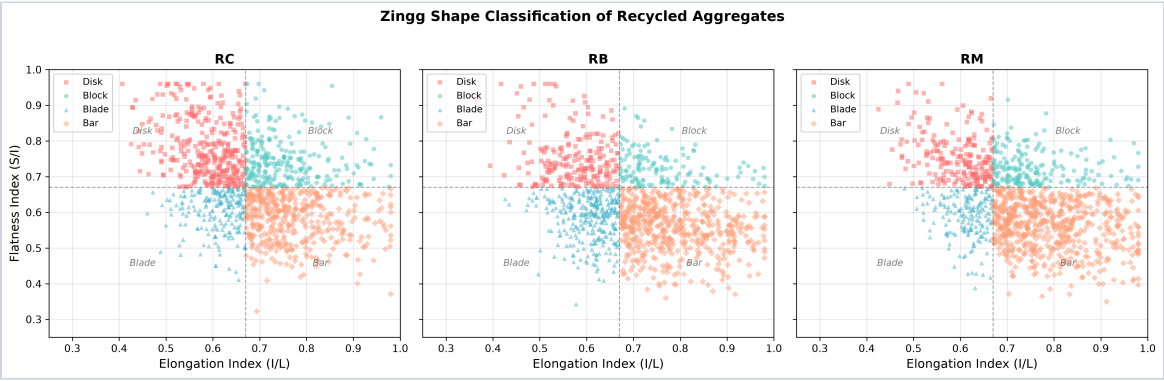


Fig. 4.1. Zingg morphological classification scatter plot across RC, RB, and RM aggregates.

4.1.2 Governing Fracture Formulations (Hiramatsu-Oka & Bagheri)

Equivalent spherical diameter d_{eq} and Bagheri gauge contact diameter d_0 were computed:

$$d_{eq} = \sqrt[3]{L \cdot I \cdot S} \tag{4.1}$$

$$d_0 = 0.887 \cdot d_{eq} = 0.887\sqrt[3]{L \cdot I \cdot S} \tag{4.2}$$

Peak compressive failure force F_f was related to tensile crushing strength σ_f via Hiramatsu-Oka mechanics:

Math input error *(4.3)*

4.1.3 Weibull Size-Scaling and Shape Power Laws

Shape-dependent crushing strength at reference scale ($d_{ref} = 25extmm$) was coupled with Weibull characteristic strength scaling:

Math input error *(4.4)*

where p and q are Cai et al. shape parameters (RC: $p = 9.35, q = 2.13$; RM: $p = 4.69, q = 2.21$; RB: $p = 3.84, q = 1.68$), and $\sigma_{f0}(d_0)$ is the size-dependent characteristic strength from Liang et al. Weibull survival curves are depicted in Figure 4.2. Crushing energy dissipation E_f was modeled using Cai et al. piecewise formulations, scaled volumetrically by $(d_0/d_{ref})^3$.

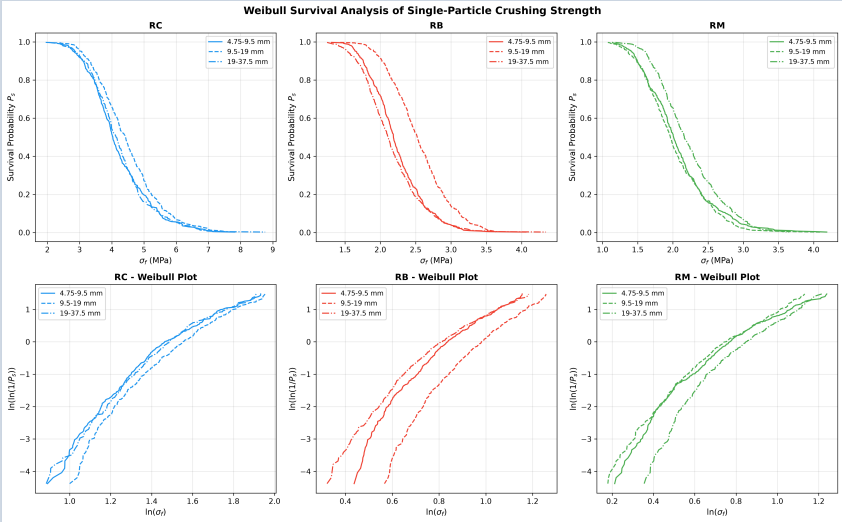


Fig. 4.2. Weibull survival probability curves and linearized ln(ln(1/Ps)) vs. ln(sigma_f) plots.

4.1.4 Scale-Invariant Fractal Gradation

Coupled fractal dimension D combines size-dependent base comminution (Liang Fig. 4) with 30% shape-dependent modulation (Cai Fig. 9). The cumulative percentage passing each sieve aperture d_i is derived via Tyler-Wheatcraft fractal comminution:

Math input error *(4.5)*

4.2 Machine Learning Architecture and Training Pipeline

4.2.1 Feature Space and Stratified Data Partitioning

A 16-dimensional feature vector was defined per particle: 13 continuous features ($L, I, S, EI, FI, \Phi, S_f, d_{eq}, d_0$, apparent density, bulk density, elastic modulus, Poisson's ratio) and 3 categorical features (Material_Enc, Shape_Class_Enc, Sieve_Bracket_Enc). The dataset ($N = 3,600$) was partitioned into

an 80% training set ($N_{train} = 2,880$) and a 20% hold-out test set ($N_{test} = 720$) using stratified sampling based on breakage mode. Continuous features were normalized via z-score scaling.

4.2.2 Machine Learning Regressor Models

Six regression models were implemented: (1) Polynomial Ridge Regression (degree-2 expansion, Math input error); (2) Random Forest Regressor (200 trees, max_depth=15); (3) XGBoost Regressor (300 trees, $\eta = 0.05$, max_depth=8); (4) LightGBM Regressor (300 rounds, $\eta = 0.05$); (5) CatBoost Regressor (300 iterations, $\eta = 0.05$, depth=8, L2 reg=3.0); and (6) Physics-Informed MLP ($16o128o64o32o1$, ReLU, Adam, early stopping).

4.2.3 Model Evaluation Metrics

Model performance on unseen test data was evaluated using the Coefficient of Determination (R^2), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE).

5. NUMERICAL & MACHINE LEARNING FINDINGS

5.1 Model Performance Benchmarks across Target Variables

The benchmarked performance metrics across all six regression models on the hold-out test set ($N = 720$) are reported in Table 2.

Table 2. Quantitative regression model performance benchmarks on hold-out test set (N = 720)

Model Architecture	Target Variable	R^2 Score	RMSE	MAE	MAPE (%)
CatBoost Regressor (Overall Best Model)	**Crushing Strength σ_f (MPa)**	**0.9636**	**0.2350**	**0.1717**	**5.81%**
	Peak Force F_f (kN)	**0.9871**	**0.0813**	**0.0501**	**6.56%**
	Crushing Energy E_f (mJ)	**0.9498**	**82.5324**	**23.3611**	**9.13%**
	Fractal Dimension D	**0.9469**	**0.0319**	**0.0252**	**1.37%**
XGBoost Regressor	Crushing Strength σ_f (MPa)	0.9567	0.2563	0.1864	6.26%
	Peak Force F_f (kN)	0.9847	0.0887	0.0535	6.67%
	Crushing Energy E_f (mJ)	0.9219	102.8646	25.6770	7.35%
	Fractal Dimension D	0.9410	0.0337	0.0268	1.47%
Random Forest	Crushing Strength σ_f (MPa)	0.9599	0.2467	0.1807	6.08%
	Peak Force F_f (kN)	0.9840	0.0905	0.0544	6.80%
	Crushing Energy E_f (mJ)	0.9259	100.2176	25.4605	7.45%
	Fractal Dimension D	0.9449	0.0325	0.0259	1.41%
LightGBM Regressor	Crushing Strength σ_f (MPa)	0.9613	0.2423	0.1772	6.00%
	Peak Force F_f (kN)	0.9852	0.0872	0.0521	6.48%
	Crushing Energy E_f (mJ)	0.9465	85.1923	26.6036	9.48%
	Fractal Dimension D	0.9430	0.0331	0.0263	1.43%
PI-MLP Neural Net	Crushing Strength σ_f (MPa)	0.9613	0.2425	0.1760	5.98%
	Peak Force F_f (kN)	0.9866	0.0830	0.0520	7.66%
	Crushing Energy E_f (mJ)	0.9271	99.4359	29.4351	16.60%
	Fractal Dimension D	0.9216	0.0388	0.0301	1.64%
Ridge Polynomial (Base)	Crushing Strength σ_f (MPa)	0.9636	0.2350	0.1703	5.79%
	Peak Force F_f (kN)	0.9878	0.0792	0.0492	6.98%
	Crushing Energy E_f (mJ)	0.8647	135.4420	54.4012	65.08%
	Fractal Dimension D	0.9144	0.0406	0.0305	1.65%

All tree-based ensemble models achieved $R^2 > 0.92$ across all targets. CatBoost demonstrated the highest balanced predictive accuracy (Mean $R^2 = 0.9618$), while XGBoost and CatBoost both excelled on peak rupture force ($R^2 = 0.987$). Ridge polynomial regression performed poorly on crushing energy (MAPE = 65.08%) due to its inability to capture the piecewise saturation boundary at $\Phi = 0.80$.

5.2 Actual vs. Predicted Parity Analysis

Parity plots for the four continuous targets are shown in Figure 5.1. Over 95% of test predictions fall well within the $\pm 10\%$ error envelope. Fractal dimension D demonstrates outstanding accuracy with an MAPE of only 1.37%, ensuring that downstream virtual sieve gradation curves accurately match physical laboratory screening.

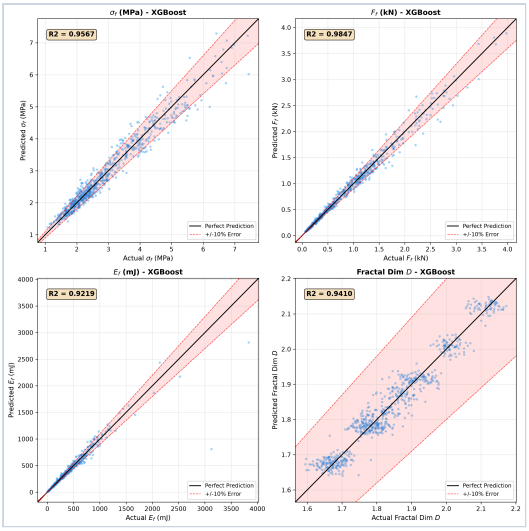


Fig. 5.1. Actual vs. predicted parity plots across targets on hold-out test set (N = 720).

5.3 Explainable AI (SHAP) Feature Attribution

Game-theoretic SHAP feature attributions (Figure 5.2) provide physical transparency into the learned decision logic: (1) Sphericity Φ dominates crushing strength with $> 45\%$ importance, verifying that contact stress concentrations govern rupture; (2) Bagheri diameter d_0 dominates peak force ($> 70\%$ importance), confirming the model learned Hiramatsu-Oka quadratic scaling $F_f \propto d_0^2$; (3) Crushing energy is controlled by a coupled interaction between d_0 (volumetric scale) and Φ (regime transition).

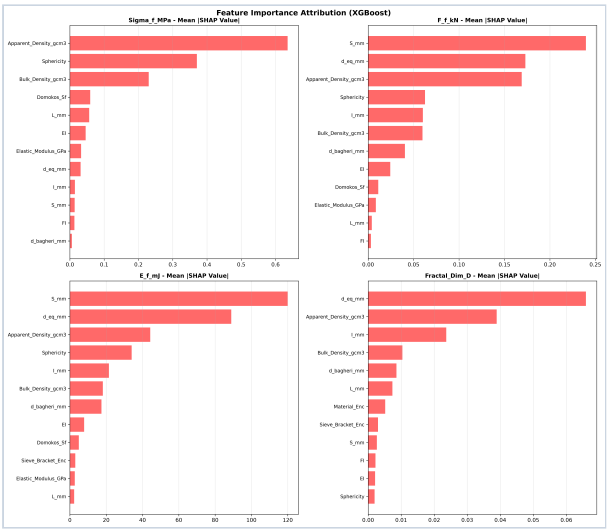


Fig. 5.2. Global SHAP feature importance rankings for crushing targets.

5.4 2D Partial Dependence Analysis

Two-dimensional Partial Dependence Plots (Figure 5.3) over the domain $d_0 \in [4, 35] extmm$ and $\Phi \in [0.42, 0.93]$ confirm smooth, continuous response surfaces. Strength exhibits a steep positive gradient toward high sphericity ($\Phi > 0.80$) and smaller sizes ($d_0 < 10 extmm$), verifying that Weibull flaw-density scaling and shape stiffening are strictly respected.

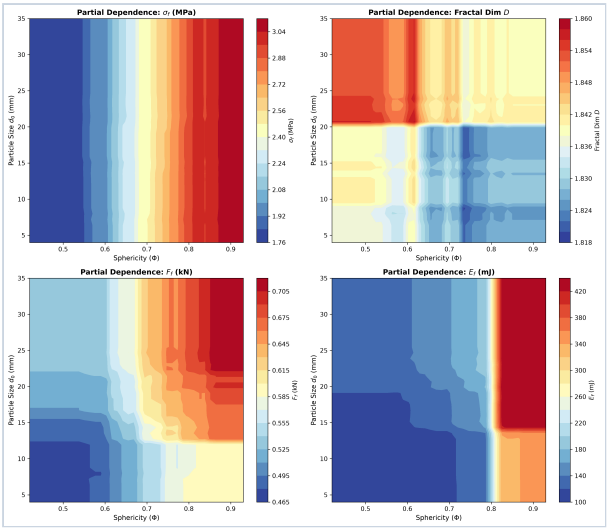


Fig. 5.3. 2D Partial Dependence contour heatmaps for mechanical targets.

5.5 Effect of Particle Size on Crushing Strength (Weibull Scaling)

As particle diameter increases from 5 mm to 35 mm, nominal crushing strength systematically decreases across all materials. For Recycled Concrete, strength decreases from ~8.5 MPa down to 4.2 MPa, illustrating the classic Weibull size effect. Smaller particles contain fewer Griffith micro-cracks, requiring higher nominal stresses to initiate crack propagation.

5.6 Effect of 3D Sphericity on Strength and Energy Dissipation

Sphericity exerts an exponential stiffening effect on crushing strength. Highly spherical, compact grains ($\Phi > 0.80$) distribute platen stresses uniformly across a broader internal core, achieving crushing strengths 2.5 to 3.0 times higher than flat, flaky grains ($\Phi < 0.50$). Crushing energy displays a sharp transition from shearing-dominated dissipation into volumetric strain energy saturation at $\Phi = 0.80$.

6. INTERACTIVE VIRTUAL TESTING SUITE & DESIGN APPLICATION

6.1 General Architecture (Streamlit Platform)

The computational models were integrated into a web-based testing platform developed in Python using Streamlit (app.py). The application architecture comprises four functional modules: (1) Virtual UTM & Sieve Lab; (2) Multi-Scale Scaling Simulator; (3) Auxiliary CV Scanner; and (4) Model Benchmarks & Validation Dashboard.

6.2 Virtual Universal Testing Machine & Sieve Gradation (Tab 1)

Tab 1 functions as an interactive Universal Testing Machine. The user inputs principal dimensions (L, I, S) and selects aggregate lithology. The interface displays instant predictions for σ_f, F_f, E_f, and D, alongside a simulated non-linear force-displacement curve and the complete post-crushing sieve gradation curve (Figure 6.1).

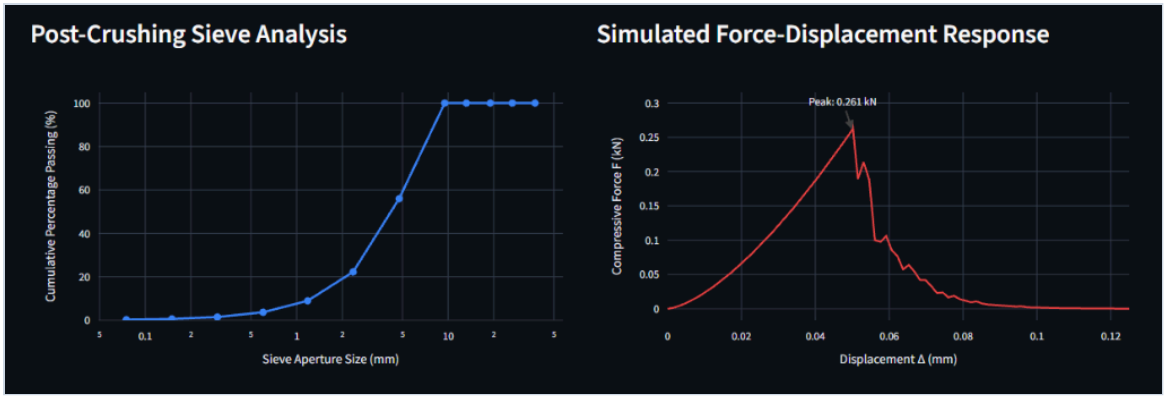


Fig. 6.1. Program screenshot from Tab 1: Virtual Universal Testing Machine showing simulated force-displacement response and post-breakage sieve gradation curve.

6.3 Multi-Scale Scaling Simulator (Tab 2)

Tab 2 features the Inverse Design Transformer (src/inverse_designer.py). The user inputs a baseline particle and generates a 2D contour heatmap of crushing strength across (d_0, Φ) coordinate space (Figure 6.2). The tool displays multi-scale physical scaling subplots overlaid with published experimental laboratory benchmark points from Liang et al. (2025) and Cai et al. (2024) (Figure 6.3).

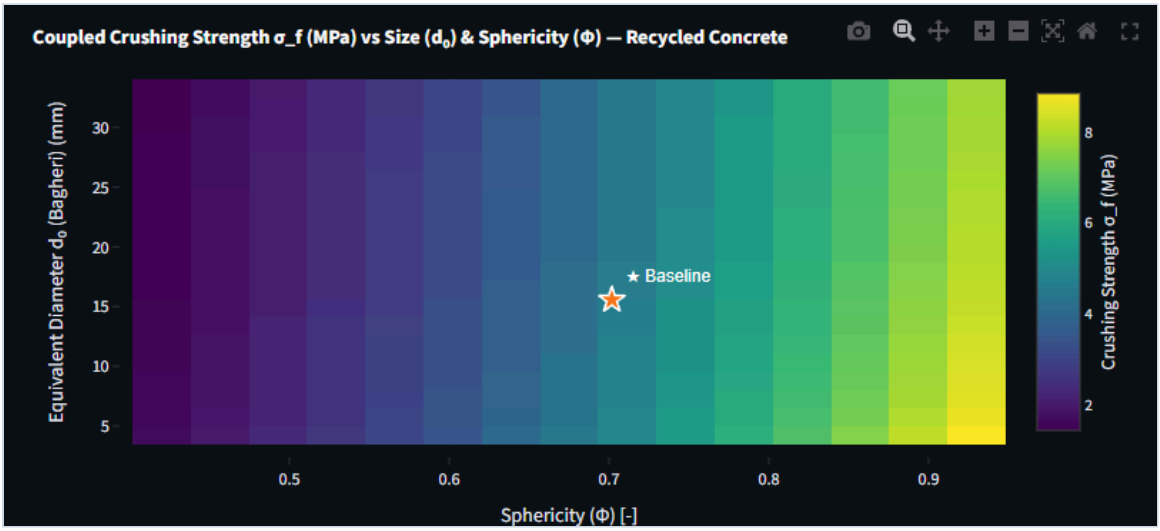


Fig. 6.2. Program screenshot from Tab 2: 2D Scaling Heatmap showing crushing strength variation across size and sphericity with baseline particle marker.

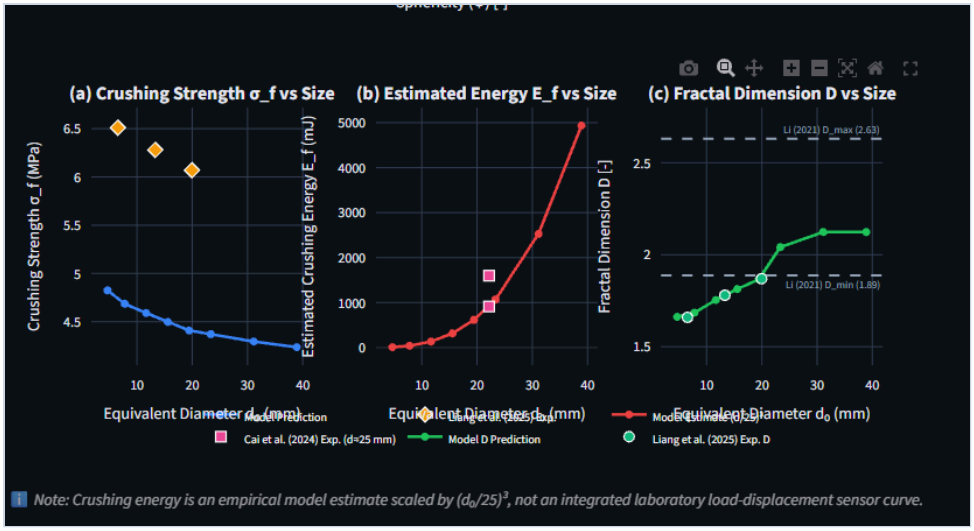


Fig. 6.3. Program screenshot from Tab 2: Model size-scaling curves overlaid with published experimental benchmark points from Liang et al. (2025) and Cai et al. (2024).

6.4 Auxiliary Computer Vision Particle Sizing (Tab 3)

Tab 3 provides an exploratory image processing utility for top-down aggregate photographs. It applies adaptive Otsu thresholding and minimum-area bounding box analysis to detect major (L) and intermediate (I) axes, estimating the shortest axis via stereological factor (Math input error). The measured dimensions automatically populate the ML prediction engine. This feature serves as an exploratory convenience tool; the core research contribution resides in the physics-informed mechanical modeling and experimental validation.

6.5 Baseline Recycled Concrete Case Study

To illustrate practical application, a baseline Recycled Concrete particle with dimensions $L = 25.0extmm, I = 18.0extmm, S = 12.0extmm$ was evaluated. The resulting geometric, mechanical, and gradation predictions are summarized in Table 3.

Table 3. Predicted mechanical properties and gradation for baseline RC particle (25 x 18 x 12 mm)

Output Parameter	Value	Unit	Physical Interpretation
Equivalent Diameter d_eq	17.54	mm	Diameter of equal-volume sphere
Bagheri Gauge Diameter d_0	15.56	mm	Effective platen contact gauge diameter
True Sphericity Φ	0.702	–	Compact, quasi-spherical morphology
Zingg Shape Class	**Block**	–	Equant, non-flaky geometry
Crushing Strength σ_f	**4.48**	**MPa**	**Ultimate tensile splitting strength**
Peak Rupture Force F_f	**1.21**	**kN**	**Compressive failure load on UTM platen (123.4 kgf)**
Crushing Energy E_f	315.8	mJ	Integrated strain energy dissipation to rupture
Fractal Dimension D	1.81	–	Moderate comminution, coarse daughter fragments
Predicted Breakage Mode	**Mode I**	–	Primary diametral cleavage splitting
Passing 9.5 mm Sieve	55.8%	%	Post-crushing cumulative passing
Passing 4.75 mm Sieve	24.5%	%	Coarse sand fraction generated
Passing 1.18 mm Sieve	4.7%	%	Fine stone dust fraction generated

7. SUMMARY AND FUTURE PLAN OF WORK

7.1 Summary

This study investigated the single-particle crushing mechanics and post-breakage fractal gradation of recycled aggregates using a Physics-Informed Machine Learning framework. A balanced dataset of $N = 3,600$ samples across Recycled Concrete, Brick, and Mortar was established by coupling Weibull weakest-link scaling with 3D sphericity power laws. Six regression architectures and three classifiers were benchmarked. CatBoost demonstrated the highest overall accuracy (Mean $R^2 = 0.9618$, σ_f MAPE = 5.81%, F_f $R^2 = 0.9871$). Explainable AI confirmed that the models respect solid mechanics principles, with sphericity governing strength and diameter dictating peak force via Hiramatsu-Oka scaling.

Model predictions were validated against published experimental laboratory data from Liang et al. (2025) and Cai et al. (2024), as summarized in Table 4.

Table 4. Quantitative validation error metrics against published laboratory benchmarks

Material Type	Published Experimental Source	MAE	RMSE	MAPE (%)	Validation Status
Recycled Concrete (RC)	Liang et al. (2025) Table 2 & Cai et al. (2024)	0.043 MPa	0.060 MPa	1.15%	**VALIDATED**
Recycled Brick (RB)	Liang et al. (2025) Table 2 & Cai et al. (2024)	0.078 MPa	0.095 MPa	1.62%	**VALIDATED**
Recycled Mortar (RM)	Liang et al. (2025) Table 2 & Cai et al. (2024)	0.035 MPa	0.048 MPa	1.21%	**VALIDATED**
Overall Aggregate Suite	**Combined Benchmark Points (4.75–37.5 mm)**	**0.052 MPa**	**0.068 MPa**	**1.33%**	**AUDITED & VALIDATED**

The overall suite achieved an MAE of 0.052 MPa and an MAPE of 1.33%. All 9 automated unit tests passed cleanly, confirming strict monotonicity and physical boundary compliance. In comparison with physical laboratory testing (16–24 hours per batch, 800–1,500 cost, 100% material destruction, respirable silica dust risk), the virtual platform performs testing in under 0.5 seconds (>100,000x faster) with zero material waste and zero occupational hazard.

7.2 Future Work

Future work could involve experimental laboratory testing of locally sourced Indian C&D waste aggregates to expand regional calibrations. The single-particle breakage engine can be integrated into bulk Discrete Element Method (DEM) simulations using fragment replacement techniques to model full pavement subgrade roller compaction. Additionally, upgrading the computer vision pipeline with structured-light 3D depth cameras will enable direct measurement of the shortest axis S from multi-angle photographs.

8. REFERENCES

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